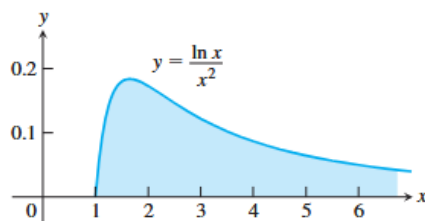
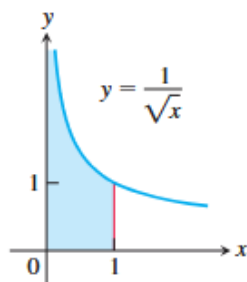


8.8 Improper Integrals

Up to now, we have required definite integrals to satisfy two properties. First, the domain of integration $[a, b]$ must be finite. Second, the range of the integrand must be finite on this domain. In practice, we may encounter problems that fail to meet one or both of these conditions. The integral for the area under the curve $y = (\ln x)/x^2$ from $x = 1$ to $x = \infty$ is an example for which the domain is infinite.



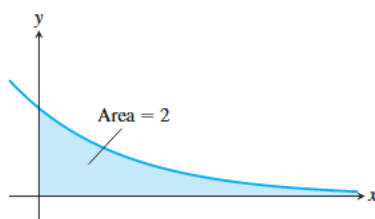
The integral for the area under the curve of $y = \frac{1}{\sqrt{x}}$ between $x = 0$ and $x = 1$ is an example for which the range of the integrand is infinite.



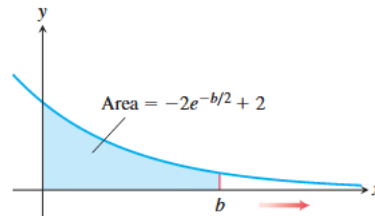
In either case, the integrals are said to be improper and are calculated as limits. We will see that improper integrals play an important role in probability. They are also useful when investigating the convergence of certain infinite series in Chapter 10.

① Infinite Limits of Integration

Consider the infinite region that lies under the curve $y = e^{-x/2}$ in the first quadrant.



You might think this region has infinite area, but we will see that the value is finite. We assign a value to the area in the following way. First find the area $A(b)$ of the portion of the region that is bounded on the right by $x = b$.



$$A(b) = \int_0^b e^{-\frac{x}{2}} dx = -2e^{-\frac{x}{2}} \Big|_0^b = -2e^{-\frac{b}{2}} + 2$$

Then find the limit of $A(b)$ as $b \rightarrow \infty$

$$\lim_{b \rightarrow \infty} A(b) = \lim_{b \rightarrow \infty} \left(-2e^{-\frac{b}{2}} + 2 \right) = 2.$$

The value we assign to the area under the curve from 0 to ∞ is

$$\int_0^{\infty} e^{-\frac{x}{2}} dx = \lim_{b \rightarrow \infty} \int_0^b e^{-\frac{x}{2}} dx = 2.$$

DEFINITION Integrals with infinite limits of integration are improper integrals of Type I.

1) If $f(x)$ is continuous on $[a, \infty)$, then

$$\int_a^{\infty} f(x) dx = \lim_{b \rightarrow \infty} \int_a^b f(x) dx.$$

2) If $f(x)$ is continuous on $(-\infty, b]$, then

$$\int_{-\infty}^b f(x) dx = \lim_{a \rightarrow -\infty} \int_a^b f(x) dx.$$

3) If $f(x)$ is continuous on $(-\infty, \infty)$, then

$$\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^c f(x) dx + \int_c^{\infty} f(x) dx$$

where c is any real number.

In each case, if the limit exists and is finite, we say that the improper integral **converges** and that the limit is the value of the improper integral. If the limit fails to exist, the improper integral **diverges**.

Note 1 The choice of c in Part 3 of the definition is unimportant.

Example 1 Is the area under the curve $y = \frac{\ln x}{x^2}$ from $x = 1$ to $x = \infty$ finite?

If so, what is its value?

Example 2 Evaluate $\int_{-\infty}^{\infty} \frac{1}{1+x^2} dx$.

Example 3 Evaluate $\int_0^{\infty} xe^{-px} dx$, where $p > 0$.

Example 4 For what values of p does the integral $\int_1^{\infty} \frac{1}{x^p} dx$ converge? When the integral does converge, what is its value?

② Integrands with Vertical Asymptotes

Another type of improper integral arises when the integrand has a vertical asymptote—an infinite discontinuity—at a limit of integration or at some point between the limits of integration. If the integrand f is positive over the interval of integration, we can again interpret the improper integral as the area under the graph of f and above the x -axis between the limits of integration.

DEFINITION Integrals of functions that become infinite at a point within the interval of integration are improper integrals of Type II.

1. If $f(x)$ is continuous on $(a, b]$ and discontinuous at a , then

$$\int_a^b f(x) dx = \lim_{c \rightarrow a^+} \int_c^b f(x) dx.$$

2. If $f(x)$ is continuous on $[a, b)$ and discontinuous at b , then

$$\int_a^b f(x) dx = \lim_{c \rightarrow b^-} \int_a^c f(x) dx.$$

3. If $f(x)$ is discontinuous at c , where $a < c < b$, and continuous on $[a, c)$ and $(c, b]$

then

$$\int_a^b f(x) \, dx = \int_a^c f(x) \, dx + \int_c^b f(x) \, dx.$$

In each case, if the limit exists and is finite, we say that the improper integral **converges** and that the limit is the value of the improper integral. If the limit fails to exist, the improper integral **diverges**.

Example 5 Investigate the convergence of $\int_0^1 \frac{1}{1-x} \, dx$.

Example 6 Investigate the convergence of $\int_{-1}^1 \frac{1}{x^2} \, dx$.

Example 7 Evaluate $\int_0^3 \frac{1}{(x-1)^{\frac{2}{3}}} \, dx$.

Example 8 For what values of p does the integral $\int_1^6 \frac{1}{(x-1)^p} \, dx$ converge? When the integral does converge, what is its value?

Example 9 Evaluate $\int_0^\infty \frac{1}{\sqrt{x(x+1)^3}} \, dx$.